From Seed to Factory: Bootstrapping and Industrial Closure for a Self-Replicating Interstellar Probe

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Working draft

Contents

Full paper — third in a series, expanded from the earlier proposal of the same title; revision 2 incorporates a pre-deposit dual review. The companion papers include the vehicle ("Growth, Not Speed"), the payload ("The Payload"), the engineering budgets (analytical and computational), the deep-time archive ("DNA Mission Ledgers"), contact governance ("Contact, Contamination, and Noninterference"), the governed-amendment problem ("Amending the Unamendable"), the subsystem budget, and the ethics of the lineage. This paper addresses the manufacturing-and-closure capability on which all of them depend: how a small landed seed becomes a factory that can mine, refine, fabricate, build instruments, make its own power and launchers, and finally produce a child probe — without resupply, oversight, or a second chance.

Abstract

The companion papers describe the *vehicle* and the *payload* of a slow, self-replicating interstellar probe but rest on a capability neither develops: the ability of a small landed seed to bootstrap itself into a full industrial base, from local asteroidal and cometary material, with no resupply. This paper addresses that capability directly. We adopt the language of **industrial closure** — materials, parts, and information closure — and argue that the realistic target is not full (100%) closure but **partial closure with a small, shrinking inventory of carried "vitamin" parts**, chiefly

advanced microelectronics, enriched fissile material, and precision metrology, which are the hardest links to close. We treat growth from seed to factory as a **staged bootstrapping cascade**, in which simple machines build more capable ones, and we render “what the probe must be able to do” as a measurable ladder of **closure levels**, from repair-only to full replication, each with its own closure fraction and bottleneck. We identify **in-situ instrument manufacture** — large telescopes and interferometers built from local optics and structure, limited mainly by detector electronics — as the capability that turns each settlement into a growing observatory. We develop, as this paper’s own quantitative contribution, the **trajectory of the vitamin mass fraction** as manufacturing capability ratchets up across the ladder and across generations: large at the first landed kit, falling toward the few-percent regime estimated in the computational engineering paper as the lineage matures, and bounded below by an irreducible vitamin set. We connect closure to the demographic condition for galactic spread — an effective reproduction number above one — by showing that a settlement reproduces only if it can close enough of its own manufacture to build both a child and its launcher. Finally, we treat the **biological-capability** question — whether the probe should synthesize DNA, and to what end — separating the broadly useful *capability* from the irreversible *act* of seeding life, and adopting the working default of **capability yes, release no**, now formalized in the governance paper. We close the triad of body, mind, and hands, and identify the single quantitative object the three papers share — since supplied at first order by the engineering companion papers.

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1. Motivation: The Capability the Vehicle and Payload Papers Defer

The vehicle and payload papers reach the same edge and stop there. The vehicle paper argues that long-term survival depends on autonomous self-repair, and concedes that high-closure self-repair — refining materials and fabricating components from raw matter — is the single largest piece of unproven technology in the architecture. The payload paper argues that a settled probe escapes the launch-mass limit and becomes a growing observatory, but only if it can *build* instruments where it lands. Both claims are underwritten by one capability: the ability of a small seed to grow into a self-sustaining factory. That capability is the crux of the entire concept. Without it, the probe is a one-shot artifact that degrades to silence; with it, the probe is the founder of an industrial lineage that can repair, observe, and reproduce indefinitely. It deserves, and here receives, its own treatment.

The idea is old and its lineage is concrete. Von Neumann’s universal constructor established the theoretical possibility that a machine can build a copy of itself, including the part of itself that does the building (von Neumann & Burks 1966). NASA’s 1980 summer study turned the abstraction into engineering, designing a self-replicating lunar factory — a “seed” that lands, mines regolith, and grows a manufacturing base — in concrete terms of mass, throughput, and machine types (NASA 1982). Freitas carried the concept to the interstellar scale, sketching a self-reproducing probe that builds copies in a target system (Freitas 1980), and the later comprehensive survey of kinematic self-replicating machines mapped the full design space, including the closure formalism this paper uses (Freitas & Merkle 2004). Most recently, Metzger and colleagues showed that the seed need not be large: on the order of twelve tonnes landed on the Moon could bootstrap a self-sustaining, self-expanding space industry over roughly two decades, given the right staging (Metzger et al. 2013); and the robotics literature has since reviewed the physical demonstrations of self-replication that ground the concept in built hardware (Moses & Chirikjian 2020). The present paper carries that lineage into the regime the earlier work did not face in full: interstellar distances, deep time, and complete autonomy, where there is no resupply, no human oversight, and the factory must eventually reproduce not just consumer goods but the probe itself.

2. The Closure Problem

We organize the analysis around *closure* — the fraction of itself a system can make for itself — in its three standard forms (Freitas & Merkle 2004). **Materials closure** is the ability to extract and refine every required element from local rock and ice. **Parts closure** is the ability to fabricate every component from those refined materials. **Information closure** is the possession of every design, process recipe, and control program needed to do so. Of the three, information closure is essentially free: designs are data, and the payload paper establishes that the probe can carry the complete manufacturing corpus and — through the archive machinery of the DNA mission-ledger paper — keep it readable across megayears. The burden therefore falls almost entirely on materials and parts — on the physical acts of refining and making, not on knowing what to refine and make.

The central thesis is that **full closure is the wrong target.** A system that can make literally one hundred percent of itself, down to the last transistor and the last reference gauge, is enormously harder to build than one that makes the overwhelming bulk of itself and carries the rest. The honest goal is therefore *partial closure*: the seed manufactures most of its own mass locally and carries only a small inventory of the parts it cannot yet make — “vitamin” parts, in the field’s term, by analogy to nutrients an organism cannot synthesize and must ingest. The research program that follows is to identify the vitamin set, estimate its mass fraction, and — the question the proposal flagged and this paper answers — ask how that fraction shrinks as the lineage’s manufacturing capability ratchets upward over generations. The mission’s deep target is to drive the carried fraction toward zero; the realistic expectation is that it falls steeply at first and then asymptotes to a small, stubborn floor. Partial closure is also where the near-term design literature has independently landed: Borgue and Hein (2021), designing a partially self-replicating probe with current technology, reach about 70% replication by mass with microelectronics carried as unreplicated stock — a measured first rung, $C \approx 0.7$, of exactly the trajectory this paper develops.

3. The Bootstrapping Cascade

A seed does not become a factory in one step; it climbs. The mechanism is a **staged ratchet** in which each rung of capability is used to build the next. The landed kit begins with the simplest possible operations — mining loose regolith, sorting and crushing it, extracting a few base materials — and uses them to fabricate simple, low-precision tools and structures. Those, in turn, build more capable machines: better furnaces, more precise mills, larger manipulators, more refined chemistry. Capability compounds, because each generation of tools widens the set of things the next generation can make. This is not speculation about exotic machinery; it is demonstrated in miniature on Earth by self-replicating rapid prototypers, whose printed parts can be assembled into further copies of the printer (Jones et al. 2011). The interstellar seed is the same principle pursued to the point where the cascade closes on the whole probe rather than on a plastic frame. Recent materials-science work extends the demonstration from clean feedstock to the feedstock a landed seed would actually have: Malekpour and Hojjati (2026) show that recycled polymer compounded with 30% lunar regolith simulant can be additively manufactured into structural lattices and functional tooling — framed there as reclaimable, sacrificial-use structures in a closed recycling loop — with regolith incorporation improving rather than degrading dimensional stability. That is direct evidence that Level 2's bulk structure and mechanical parts (Section 4) are reachable with genuinely local, non-exotic feedstock rather than pre-refined printer stock.

Additive manufacture holds a load-bearing place in this cascade for a structural reason, not a fashionable one. Every other formative process needs part-specific tooling — casting needs moulds, forging needs dies, machining needs fixtures — and each of those must itself be made, which is exactly the recursion the ratchet exists to climb. A printer replaces part-specific tooling with part-agnostic geometry: one general-purpose forming machine per material class, directed by a file rather than a die, so much of the tooling tree collapses into information — the one input the seed carries at negligible mass. The trade has honest bounds. Printing does not remove subtractive finishing, since tolerances and surface finish still demand it; its feedstock preparation is a process chain of its own; and — the binding condition at seed scale — it must run without a quality organization. A systematic review of AI-enabled quality control for directed energy deposition, across 145 studies, finds in-process defect detection, melt-pool monitoring, and early-warning models well developed while direct measurement, validation, and qualification still rest on conventional inspection (Debnath et al. 2026). For the seed, that division of labour is the design requirement in miniature: the printer's in-process monitoring must mature into closed-loop control, and the qualification step terrestrial practice still assigns to conventional methods is what the subsystem budget paper's P4 metrology stage exists to carry.

A consequence for the architecture, following the companion papers, is that the **child probe is not a copy of the mature factory.** It is the smallest system able to survive the journey, brake, land, begin extraction, and *restart the cascade* — a mobile bootstrap package rather than a packed industrial city. The factory is heavy, fixed, and grown in place; what travels is the seed of a factory. This is why the launch problem and the closure problem are linked: the lighter the bootstrap package can be made while still able to climb the cascade unaided, the cheaper it is to launch and brake, and the more readily a settlement can afford to make and send one. The vehicle paper's stationary-settlement architecture and this paper's mobile-seed architecture are two halves of the same design: the lineage advances by growing a factory, shedding a seed, and growing another factory one star onward.

4. The Closure-Level Ladder

"What the probe must be able to do" is best expressed not as a single capability but as a measurable hierarchy, each level a milestone with its own closure fraction and its own binding bottleneck. We define five levels:

- **Level 1 (L1) — Repair from spares.** The probe replaces failed components from a carried inventory but manufactures nothing. Closure is essentially zero; survival time is set by the spares budget and fails over deep time. This is the floor the vehicle paper showed to be insufficient on its own.
- **Level 2 (L2) — Bulk structure and mechanical parts.** The probe mines, refines common metals and silicates, and fabricates structural and mechanical components — frames, tanks, mirror blanks, simple mechanisms. Closure rises sharply by mass, because structure is most of the mass; the bottleneck is anything requiring precision or exotic materials.
- **Level 3 (L3) — Scientific instruments.** The probe manufactures optics, mirrors, and the mechanical bulk of telescopes and interferometers from local material, carrying only detectors, control electronics, and precision actuators. This is the level that turns a settlement into an observatory (Section 5).
- **Level 4 (L4) — Power and launch infrastructure.** The probe builds its own reactor structure, radiators, power-handling, and the fixed launch infrastructure — mass drivers, beam directors, or solar-thermal launchers — that sends a child on its way, together with the coarse, high-volume electronics that power-handling and routine control require (Section 6). Fissile closure begins, at best, late in this level and completes only at Level 5; until then fissile material and high-performance electronics remain carried or hard-won.
- **Level 5 (L5) — Full self-replication.** The probe closes the hardest links and reproduces itself entirely from local material, including the microelectronics, the fissile inventory, and the metrology that anchors everything else. Closure approaches one; the vitamin set approaches zero. This is the level at which the lineage becomes truly resupply-independent.

The ladder is a design tool, not a claim that all five are equally near. Levels 1–3 are extrapolations of demonstrated terrestrial and proposed lunar capability; Level 4 is harder but conceivable; Level 5 is the research frontier, and whether it is reachable at all is precisely the question the vitamin-set analysis (Sections 6–7) is meant to bound. A mission need not reach Level 5 to be worthwhile — a lineage stalled at Level 4, carrying a small fissile-and-chip vitamin set forward from settlement to settlement, can still expand and still learn — but Level 5 is the only level at which expansion is unconditionally self-sustaining.

5. In-Situ Instrument Manufacture

The scientific payoff of closure is concentrated at Level 3, and it is large. A telescope is, by mass, overwhelmingly structure and optics: the mirror substrate, the supporting truss, the baffles, the mounts, the mechanisms. All of these are makeable from local silica and metal, the same materials the bulk-fabrication level already handles. What is *not* locally makeable, at first, is the small high-value core — the detector arrays, the readout and control electronics, the precision actuators. Instruments therefore split cleanly into a locally manufacturable majority and a small electronic vitamin set, and this split is what lets observing power grow with the settlement rather than being fixed at launch. A probe that arrives carrying a modest detector package can, over time, build the large collecting apertures and long interferometric baselines that the detector package feeds, turning a launch-mass-limited instrument into a settlement-limited one. The payload paper's claim that a settled probe becomes a growing observatory rests entirely on this: not that the probe carries great instruments, but that it carries the *seeds* of great instruments and grows them in place. The same logic extends to every tool of exploration and manufacture — the bulk is local, the brains are carried — and it is the recurring shape of the whole closure problem in miniature.

6. The Three Bottlenecks

The feasibility of the entire scheme rests on a small number of hard links, because they determine the vitamin mass fraction and hence whether closure can be driven low enough to matter. Three dominate.

Microelectronics and computing substrates. The probe's mind, its sensors, and its detectors all depend on high-performance semiconductors, whose manufacture on Earth requires an enormous, precise, and chemically exotic supply chain. Closing this link in situ is the hardest single problem in the architecture. There are two routes, and they trade against each other. The first is to manufacture *coarse but adequate* electronics locally — larger feature sizes, simpler processes, lower performance — accepting that a settlement-made computer is far worse than a carried one but is plausibly buildable from local materials: the route Ellery (2016) develops in the lunar context, arguing that 3D-printed actuators and control electronics from regolith-derived feedstock would constitute an existence proof of universal construction. The second is to carry high-performance chips as the central vitamin, accepting a permanent dependence on an inventory that cannot be replenished and must be hoarded, rationed, and protected across deep time. The realistic architecture is likely a blend: carry the high-performance core that the mind and the best detectors need, and close the coarse, high-volume electronics that structure, power-handling, and routine control require. How small the carried core can be made is the dominant uncertainty in the whole closure budget.

Fissile material. The vehicle paper and the engineering papers establish that only a fission reactor is mass-sane for the probe's power needs over centuries, and fissile material is not something a young settlement can casually make. Enrichment requires isotope separation at industrial scale; breeding requires a reactor already running. A settlement can plausibly extract uranium from local material in principle, but concentrating and enriching it to reactor grade is a Level-4-to-5 capability, and until then the fissile inventory is carried, rationed, and treated as a vitamin as precious as the chips. The coupling to launch is direct: a child must be sent with enough fissile material to run until its own settlement can close the link, which sets a hard floor on the bootstrap package mass.

Precision metrology. The least obvious bottleneck is the most foundational. Everything a factory makes is only as accurate as the references against which it measures — the gauges, standards, and calibrations that define what “to spec” means. A self-replicating factory must therefore reproduce not only its tools but the *standards that calibrate its tools*, and metrology has a tendency to degrade across copying generations exactly as information does without error-correction: a factory built by a slightly-off factory is slightly-off-er, and the error compounds. Closing the metrology link means carrying or reconstructing absolute physical references — anchored ultimately in atomic transitions and fundamental physical constants, which are the same everywhere and can in principle be re-derived from physics rather than copied from a master gauge — the same move the 2019 redefinition of the SI made on Earth, fixing every base unit to defining constants rather than artifacts (BIPM 2019). Whether a settlement can rebuild its metrology from first principles, rather than carrying a master set of references forward forever, is a genuine open question and a quiet determinant of whether Level 5 is reachable.

7. The Vitamin Fraction Over Time

The proposal promised a quantitative treatment of how the vitamin mass fraction varies with manufacturing capability, and that is this paper’s own contribution to the series’ shared budget. We state it as a trajectory, not a point, and we are deliberately order-of-magnitude rather than falsely precise, because the underlying numbers are uncertain by factors of several.

At the first landed kit (Level 1–2), the carried fraction is large: the seed can make structure but little else, so a substantial share of its high-value mass — electronics, detectors, fissile fuel, reference instruments — is vitamin that had to be brought. This starting point is no longer only qualitative: the subsystem budget paper prices the reference seed’s carried high-value core per component, and Borgue and Hein’s (2021) near-term design measures it independently at roughly 30% of probe mass. As the cascade climbs, the fraction falls steeply, because the things that close first (structure, bulk metals, optics, simple mechanisms) are most of the mass, so closing them retires most of the carried-mass burden quickly. By the mature manufacturing level the computational engineering paper’s point estimate applies: a vitamin fraction of order a few percent — ~30 kg per tonne — dominated by the three bottlenecks of Section 6. Below that the trajectory flattens, because what remains is the irreducible set: the high-performance chips, the fissile core, and the metrology references that close last if they close at all. Across generations the floor can ratchet lower still, as each settlement that closes one more link sends children that no longer need to carry it; but the floor is bounded below by whichever of the three bottlenecks the lineage never manages to close, and a lineage that closes none of them is not self-sustaining no matter how much structure it can make.

Two implications follow. First, the design lever is not total closure but the *bottleneck set*: the entire feasibility question reduces to how light the carried core of chips, fissiles, and references can be made, because everything else closes relatively early and cheaply. Second, closure is not a property a probe has or lacks but a trajectory it climbs, and a mission can be valuable at any height on that trajectory provided the carried vitamin fraction is small enough to launch — which connects this paper’s budget directly to the demographics of expansion.

8. The Central Design Question: Biological Capability

The most consequential open question is whether the probe should be able to **synthesize DNA and engineer biology**, and to what end. The proposal separated two things usually conflated, and the governance paper has since formalized the distinction into a hard rule; we restate it here as it bears on manufacturing.

The **capability** — synthesizing nucleic acids and engineering organisms — is broadly useful on three non-controversial grounds. First, **data storage**: DNA is the densest and among the most durable information media known (Church et al. 2012; Goldman et al. 2013), the natural substrate for the deep-time archive whose density and durability figures are developed in the DNA mission-ledger paper, and information can even be encoded into the genomes of living, dividing cells that copy it as they reproduce (Shipman et al. 2017). Second, **manufacturing**: engineered biology performs nanoscale self-assembly and chemical processing "for free," a powerful complement to mechanical fabrication that may help close exactly the fine-scale links that defeat coarse machinery. Third, **preservation**: synthesis capability turns a stored genome from a record into a *re-constitutable backup* of Earth's biosphere, able to be rebuilt rather than merely read.

The **act** of using that capability to seed living organisms into a planetary environment — directed panspermia (Crick & Orgel 1973) — is a different matter entirely, and it is the genuine value fork. It would convert the probe from the observer-and-librarian of the payload paper into an active agent that spreads life, with arguments cutting hard both ways: life's continuation and proliferation as a positive good and an insurance against extinction, versus the contamination of pristine or possibly inhabited worlds, the irreversibility of release, and the hazard of acting on a founding judgment that cannot be revisited across the light-millennia separating a probe from anyone who might object. This paper does not adjudicate that fork; the governance paper does, and adopts exactly the working position proposed here: **capability yes, release no.** The probe develops the ability to synthesize DNA for storage, manufacturing, and biosphere backup, but the release of self-reproducing organisms into a world is a separate act, disabled by default in the immutable core and enabled only under stringent, simultaneously required, ledger-recorded conditions on a confirmed-sterile target. Manufacturing is thus permitted to build the biological *tools* while being forbidden, by default, to deploy them as biological *agents* — the same capability turned inward for memory and fabrication rather than outward onto a world. We note, as the proposal did, that whether distant descendants should ever be able to change that default is the governed-amendment problem — opened by the payload paper, flagged here, and since taken up in full by the governed-amendment paper ("Amending the Unamendable").

9. Coupling to Expansion

Manufacturing closure is not an end in itself; it feeds the demographic condition for galactic spread developed in the analytical and computational engineering papers, where expansion requires an effective reproduction number above one. The link is direct and sobering. A settlement reproduces only if it can close enough of its own manufacture to build a child *and* the launcher that sends it — and closure failure is one of the terms that pushes the reproduction number below one. A lineage that can repair itself but cannot build a launcher does not expand; a lineage that can build a child but must endow it with too large a vitamin inventory may find each child too heavy to launch or too dependent to survive. “Can it build a child” (this paper’s closure-and-bootstrapping question) and “do enough children succeed” (the question the vehicle paper posed and the engineering papers now compute) are therefore the two halves of replication feasibility, and the computational engineering paper’s headline knife-edge — that the reproduction number sits near one and is acutely sensitive to per-stage reliability — is in large part a statement about closure: a settlement that closes more of its manufacture makes children more cheaply, more reliably, and more often, pushing the reproduction number up. One that closes less makes the whole lineage extinction-prone. Closure is, in this sense, the engine under the demographics.

10. Working Positions and the Completed Triad

This paper has argued four working positions, each stated in the proposal as a hypothesis and here defended. That **partial closure with a shrinking vitamin set** is the right target rather than full closure, because the carried fraction falls steeply once structure closes and the only question that matters is how small the irreducible core can be made. That the **vitamin set is dominated by microelectronics, enriched fissile material, and precision metrology**, the three links that close last if they close at all. That **instruments are mostly locally manufacturable** save for detectors and control electronics, which is what lets observing power grow with the settlement. And that **biological capability should be built but biological seeding gated off by default**, the position the governance paper has since made a hard rule. None of these is a finished result; each is a defensible architecture whose first-order quantitative form the engineering papers have since supplied — what remains open is the full engineering model, and the ladder’s upper rungs.

Two further caveats belong here because they arise from this paper’s own mechanisms rather than from the value-corruption questions the ethics paper’s discussion of the lineage has since taken up. First, the metrology bottleneck of Section 6 is not only the hardest closure problem but the strangest one: if a factory’s reference standards drift and the next factory it builds inherits that drift, the error compounds silently, with no master gauge anywhere in the lineage against which the descendant could ever detect its own deviation, and no factory, seed, or settlement anywhere in the chain that did anything wrong. That marks it as categorically distinct from the deliberate corruption of the immutable core that the governed-amendment paper treats — this is an engineering-negligence problem passed down generations with no moral-agency structure at all, harm without a wrongdoer, and whether it can be bounded turns entirely on whether metrology can be re-derived from physical constants (Section 6) rather than copied forward from a carried master set. Second, the closure-level ladder of Section 4 states no priority rule for a capability-constrained settlement choosing between Level 3 (observatory and instrument capability) and Levels 4–5 (launch infrastructure and reproduction), and Section 9’s coupling to expansion shows why the silence matters: a settlement that races to close the reproduction-and-launch links ahead of the instrument link is, in effect, deprioritizing the “carry knowledge forward” purpose the series treats as primary in favor of raising its own reproduction number. Nothing in this paper, or in the series' account of internal allocation more broadly, offers a principle for making that trade-off; it is left as a policy each settlement sets by default rather than by design.

With this paper the triad closes. The vehicle paper is the *body* — propulsion, power, braking, replication. The payload paper is the *mind* — cognition, memory, mission. This paper is the *hands* — the manufacturing and closure that let the body repair and reproduce, and let the mind build the instruments through which it learns. The three shared one deferred quantitative object, approached from three sides and since taken up: whether enough children succeed (posed by the vehicle paper, computed by the engineering papers), whether a child can build the next factory (this paper's closure-and-bootstrapping question, budgeted by the engineering papers and priced per component by the subsystem budget paper), and whether the mission survives copying (the payload paper's information-integrity question, given its substrate by the DNA mission-ledger paper). A probe that cannot make its own hands cannot keep its own body alive or grow its own mind's instruments; closure is the capability on which the other two depend, and closing it — in fact, not just in design — is the research program this series exists to open.

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